

JUSTIN TRAN

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Education

University of California, Santa Cruz

Sep. 2024 – June 2028

Bachelor of Science in Computer Science — Dean's List — GPA: 3.92/4.0

Santa Cruz, California

Experience

Software Developer Intern

June 2026 – Present

IBM

Durham, North Carolina

- Replaced a 12-step manual artifact-transfer process with an **automated CI/CD pipeline** in Jenkins and Groovy, **adopted by 6 of 8 developers** on the NetView team
- Built an **AI triage pipeline (Node.js + IBM WatsonX)** that filters Mend vulnerability scan findings into true/false positives, auto-creating Jira tickets and draft PRs only for confirmed issues
- Automated Zowe CLI regression testing via a Java/Maven + YAML framework covering **13 of 15 REST endpoints across 38 scenarios**, replacing manual execution with a single make command
- Cut failure triage from **10–15 minutes to instant** by generating reports that correlate failing tests with git commit history and changed source files, with a ready-to-paste AI prompt

Software Engineer Intern

Jan. 2026 – May 2026

Rekrut AI

San Francisco, California

- Integrated a **fine-tuned open-source LLM** into **Supabase** backend to power automated skill assessments and grading, reducing manual candidate screening time
- Designed responsive, data-driven dashboards using **React.ts** to visualize key hiring metrics and candidate analytics, enabling recruiters to make actionable, bias-free decisions
- Optimized **database schema and API endpoints** to support Intelligent Profile Matching algorithms, improving the accuracy of candidate-to-role recommendations

Projects

matchbox – Matching Engine | C++, Kafka, Docker, Kubernetes, Terraform, AWS (EKS), Prometheus/Grafana **2026**

- Built a C++ limit-order-book matching engine sustaining **7.7M orders/sec at 83ns p50 / 375ns p99**, using sorted price-level maps with an order-ID index for **O(1) top-of-book and cancels**
- Implemented price-time and pro-rata matching behind one strategy interface, **benchmarking both on identical 500K-action streams to trace a 6.6× throughput gap to 439-maker order splits**
- Simulated informed vs. noise traders across 12 concurrent symbols, **improving price discovery 4.6× (mean |trade – fair value|: 120 → 26 ticks)**, with thread isolation proven by ThreadSanitizer
- Deployed 4 symbol-sharded replicas on Kubernetes/EKS via Terraform and Kafka**, verifying **11,606** trades matched ground truth exactly at **1,455 orders/sec, 2.34µs p99** via Prometheus/Grafana

NextPlay – NBA Prediction App | React, Tailwind CSS, Django, MySQL, Pandas, scikit-learn

2025

- Built a full-stack ML app (Django REST, React, scikit-learn) predicting NBA player performance from **15,000+ game records, with automated daily game ingestion and model retraining**
- Trained a **Random Forest regression model** with custom feature engineering including rolling averages and contextual metrics, deploying via REST API with JWT authentication and MySQL database
- Built an **automated data pipeline** that fetches, validates, and ingests daily NBA statistics, re-trains updated models, and publishes to Hugging Face Hub for version control, eliminating manual workflows

InterviewBot – Tailored Mock Interviews | React, Django, OpenAI API, Tailwind CSS

2025

- Engineered full-stack mock interview platform with React frontend and Django REST API, integrating **OpenAI** to generate tailored interview questions; deployed for private use among **80+ users** at UCSC's ACM club
- Designed **RESTful API** architecture handling asynchronous AI requests, JSON parsing, and error handling; serves dynamic interview content analyzing user responses across clarity, relevance, and sentiment metrics
- Deployed full-stack application** to Railway + Vercel with production configurations, managing environment variables and CORS settings for secure cross-origin communication

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, C, C++, C#, Shell, Groovy

Technologies/Frameworks: React, Django, Apache Kafka, Node.js, Supabase, MySQL, OpenAI API, Tailwind CSS

Cloud & DevOps: AWS (EKS, EC2, VPC, ECR), Docker, Kubernetes, Terraform, Helm, Prometheus, Grafana, Jenkins

Libraries: Pandas, NumPy, scikit-learn, librdkafka, Catch2

Developer Tools: VS Code, Git, GitHub, Jenkins, Hugging Face, CMake, kubectrl, ThreadSanitizer